

Table 1: Published calibrations between sedimentary leaf wax $\delta^2\text{H}$ and precipitation $\delta^2\text{H}$. Global calibrations are shown first, followed by regional studies organized by continent. Slope and intercept refer to the linear relationship $\delta^2\text{H}_{\text{wax}} = (\text{Slope} \times \delta^2\text{H}_{\text{precip}}) + \text{Intercept}$. Some studies report relationships with other variables (e.g., aridity index, elevation, lake water enrichment) as noted.

Study	Region	Compound	Slope	Intercept	R^2	n	Error	Notes
<i>Global Calibrations</i>								
Sachse et al. 2012	Global	n-C ₂₉ alkane	0.548	−148	0.80	NR	–	Lake sediments
McFarlin et al. 2019	Global	n-C ₂₉ alkane	0.78	−122	0.80	177	RSE=16	Lake sed. vs annual precip
McFarlin et al. 2019	Global	n-C ₂₉ alkane	0.82	−142	0.80	177	RSE=20	Lake sed. vs summer precip
Ladd et al. 2021	Global	n-C ₂₉ alkane	0.92 ± 0.02	−117 ± 2	0.74	665	–	Soils + lake sediments
Gaviria-Lugo et al. 2023	Global	n-C ₂₉ alkane	0.79 ± 0.02	−124 ± 1	0.71	750	–	Soils + lakes, 26 studies
Gaviria-Lugo et al. 2023	Global	n-C ₃₁ alkane	0.76 ± 0.02	−130 ± 1	0.73	663	–	Soils + lakes, 26 studies
Douglas et al. 2012	Global	n-C ₂₉ alkane	0.86	−156	0.80	NR	–	Compilation incl. SE Mexico
McFarlin et al. 2019	Global	n-C ₂₈ acid	0.70	−130	0.70	212	RSE=24	Lake sed. vs annual precip
McFarlin et al. 2019	Global	n-C ₂₈ acid	0.73	−134	0.70	212	RSE=24	Lake sed. vs summer precip
Ladd et al. 2021	Global	n-C ₂₈ acid	0.73 ± 0.03	−114 ± 3	0.74	242	–	Soils + lake sediments
Douglas et al. 2012	Global	n-C ₂₈ acid	0.81	−108	0.63	NR	–	Compilation incl. SE Mexico
McFarlin et al. 2019	Global	n-C ₂₄ acid	0.70	−121	0.70	123	RSE=23	Lake sed. vs annual precip
McFarlin et al. 2019	Global	n-C ₂₄ acid	0.60	−145	0.70	110	RSE=23	Lake sed. vs lake water

Continued on next page

Table 1 – continued from previous page

Study	Region	Compound	Slope	Intercept	R ²	n	Error	Notes
McFarlin et al. 2019	Global	n-C ₂₃ alkane	0.75	−121	0.30	68	RSE=34	Lake sed. vs annual precip
McFarlin et al. 2019	Global	n-C ₂₃ alkane	0.55	−159	0.50	44	RSE=29	Lake sed. vs lake water
Guenther et al. 2013	Global	n-C ₂₃ alk. (aquatic)	0.95	−147.5	0.76	~35	–	vs lake water
Guenther et al. 2013	Global	n-C ₂₉ alk. (terrestrial)	0.87	−127.6	0.81	~35	–	vs surface water
<i>Regional Calibrations – North America</i>								
Hou et al. 2008	SW USA (entire)	C ₂₈ acid	0.93	−99.1	0.76	32	RMS=122	Sediments vs $\delta^2\text{H}_p$
Hou et al. 2008	SW USA (Basin & Range)	C ₂₈ acid	1.69	−42	0.84	18	RMS=70	<400mm precip
Hou et al. 2008	SW USA (Great Plains)	C ₂₈ acid	0.67	−109.7	0.64	14	RMS=61	>400mm precip
Huang et al. 2004	Eastern USA	n-C ₂₈ acid	0.423	−144	0.38	NR	–	vs lake water
Polissar & Freeman 2010	N+S Am. (tropical forest)	n-C ₂₉ alkane	0.2 ± 0.8	−125 ± 5	0.08	5	–	ε_{app} vs lake water enrich.
Polissar & Freeman 2010	N Am. (temp. alpine forest)	n-C ₂₉ alkane	0.4 ± 2.3	−98 ± 6	0.89	10	–	ε_{app} vs lake water enrich.
Polissar & Freeman 2010	Venezuela (páramo shrub.)	n-C ₂₉ alkane	1.8 ± 0.9	−120 ± 9	0.64	11	–	ε_{app} vs lake water enrich.
<i>Regional Calibrations – Greenland</i>								
McFarlin et al. 2019	Greenland	n-C ₂₄ acid	0.91	−112	0.70	21	RSE=24	vs annual precip
McFarlin et al. 2019	Greenland	n-C ₂₄ acid	0.39	−185	0.02	21	RSE=31	vs lake water
McFarlin et al. 2019	Greenland	n-C ₂₃ alkane	0.80	−115	0.80	21	RSE=14	vs annual precip
McFarlin et al. 2019	Greenland	n-C ₂₃ alkane	0.44	−175	0.30	21	RSE=26	vs lake water
<i>Regional Calibrations – South America</i>								
Feakins et al. 2016	Andes-Amazon, Peru	C ₂₉ alk + C ₂₈ acid	0.89	−134	0.97	9	–	Plot means vs stream water
Feakins et al. 2016	Andes-Amazon, Peru	C ₂₉ alkane	0.79	−132	0.43	NR	–	Individuals vs xylem water

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Table 1 – continued from previous page

Study	Region	Compound	Slope	Intercept	R ²	n	Error	Notes
Feakins et al. 2016	Andes-Amazon, Peru	C ₃₁ alkane	0.65	−147	0.42	NR	–	Individuals vs xylem water
Feakins et al. 2016	Andes-Amazon, Peru	C ₂₄ acid	0.95	−110	0.48	NR	–	Individuals vs xylem water
Feakins et al. 2016	Andes-Amazon, Peru	C ₂₆ acid	1.00	−111	0.54	NR	–	Individuals vs xylem water
Feakins et al. 2016	Andes-Amazon, Peru	C ₂₈ acid	0.83	−128	0.38	NR	–	Individuals vs xylem water
Feakins et al. 2016	Andes-Amazon, Peru	C ₃₀ acid	0.71	−135	0.14	NR	–	Individuals vs xylem water
Feakins et al. 2016	Andes-Amazon, Peru	C ₂₇ alkane	0.54	−150	0.27	NR	–	Individuals vs xylem water
Feakins et al. 2016	Andes-Amazon, Peru	C ₂₈ acid	−18 km ^{−1}	−151	0.26	102	–	Individuals vs elevation
Feakins et al. 2016	Andes-Amazon, Peru	C ₂₉ alkane	−14 km ^{−1}	−163	0.34	136	–	Individuals vs elevation
Feakins et al. 2016	Andes-Amazon, Peru	C ₂₈ acid (plot means)	−20 km ^{−1}	−161	0.97	4	–	vs elevation (outlier removed)
Feakins et al. 2016	Andes-Amazon, Peru	C ₂₉ alk. (plot means)	−18 km ^{−1}	−164	0.97	5	–	vs elevation
<i>Regional Calibrations – Central America & Mexico</i>								
Douglas et al. 2012	SE Mexico & C. Am.	n-C ₂₈ acid	−19*	−96	0.25	~39	–	$\varepsilon_{\text{wax/w}}$ vs AI
Douglas et al. 2012	SE Mexico & C. Am.	n-C ₂₉ alkane	−21*	−95	0.48	~36	–	$\varepsilon_{\text{wax/w}}$ vs AI
Douglas et al. 2012	SE Mexico (lakes)	n-C ₂₈ acid	−25*	−89	0.58	~16	–	$\varepsilon_{\text{wax/w}}$ vs AI
Douglas et al. 2012	SE Mexico (lakes)	n-C ₂₉ alkane	−28*	−85	0.56	~16	–	$\varepsilon_{\text{wax/w}}$ vs AI
Douglas et al. 2012	SE Mexico (lakes)	n-C ₃₀ acid	−38*	−67	0.70	~16	–	$\varepsilon_{\text{wax/w}}$ vs AI
<i>Regional Calibrations – Africa</i>								
Vogts et al. 2016	SW Africa	C ₂₉ alkane	–	–	0.64	9	–	Marine sed., r=0.80 vs Dec precip

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Table 1 – continued from previous page

Study	Region		Compound	Slope	Intercept	R ²	n	Error	Notes
Vogts et al. 2016	SW Africa		C ₃₁ alkane	–	–	0.45	9	–	Marine sed., r=0.67 vs annual precip
Herrmann et al. 2016	S. Africa (SRZ)		WMA C ₂₉ +C ₃₁	–	–	0.42	21	–	Soils, r=0.65 vs annual precip
Herrmann et al. 2016	S. Africa (WRZ)		WMA C ₂₉ +C ₃₁	–	–	0.23	21	–	Soils, r=–0.48 vs annual precip
Herrmann et al. 2016	S. Africa (YRZ)		WMA C ₂₉ +C ₃₁	–	–	0.67	6	–	Soils, r=0.82 vs annual precip
<i>Regional Calibrations – Asia (Tibet)</i>									
Guenther et al. 2013	Tibetan (E-W)	Plateau	n-C ₂₃ alk. (aquatic)	1.95	–47.0	0.89	6	–	vs lake water
Guenther et al. 2013	Tibetan (E-W)	Plateau	n-C ₂₉ alk. (terrestrial)	0.81	–183.4	0.72	6	–	vs surface water
Guenther et al. 2013	Tibetan (combined)	Plateau	n-C ₂₃ alk. (aquatic)	1.89	–46.3	0.94	~12	–	vs lake water, 4 studies
Guenther et al. 2013	Tibetan (combined)	Plateau	n-C ₂₉ alk. (terrestrial)	0.70	–129.2	0.66	~13	–	vs surface water, 4 studies
<i>Regional Calibrations – Middle East</i>									
Goldsmith et al. 2019	Israel		Avg C ₂₅ –C ₃₅	1.25	–117	0.65	12	±7‰	vs rainfall δ ² H
Goldsmith et al. 2019	Israel		Avg C ₂₅ –C ₃₅	0.8 [†]	–66.4	0.73	12	6.9‰	Multiple regr. incl. RH
<i>Plant Functional Type Regressions (Living Plants, Global)</i>									
Sachse et al. 2012	Global		n-C ₂₉ alk. (Trees)	0.524	–134	0.46	99	–	Living plant tissues
Sachse et al. 2012	Global		n-C ₂₉ alk. (Shrubs)	0.867	–112	0.21	79	–	Living plant tissues (n.s.)
Sachse et al. 2012	Global		n-C ₂₉ alk. (Forbs)	1.158	–120	0.83	27	–	Living plant tissues
Sachse et al. 2012	Global		n-C ₂₉ alk. (C ₃ gram.)	1.209	–129	0.49	39	–	Living plant tissues
Sachse et al. 2012	Global		n-C ₂₉ alk. (C ₄ gram.)	0.777	–142	0.48	56	–	Living plant tissues

Abbreviations: NR = Not Reported; RSE = Residual Standard Error; RMS = Root Mean Square; WMA = Weighted Mean Average; SRZ/WRZ/YRZ = Summer/Winter/Year-round Rainfall Zone; n.s. = not significant; alk. = alkane; gram. = graminoids; sed. = sediments; temp. = temperate; enrich. = enrichment; regr. = regression; AI = Aridity Index.

*Slope vs Aridity Index (dimensionless), not vs $\delta^2\text{H}_{\text{precip}}$. Douglas et al. (2012) found no significant direct relationship between $\delta^2\text{H}_{\text{wax}}$ and $\delta^2\text{H}_{\text{precip}}$ in this region.

[†]Multiple regression coefficient for $\delta^2\text{H}_{\text{precip}}$ term; model also includes relative humidity term with coefficient -0.9 .